Finite-horizon LQR in Simulink (Kirk Examples 5.2-1 and 5.2-2)

Optimal Control, Homework 4

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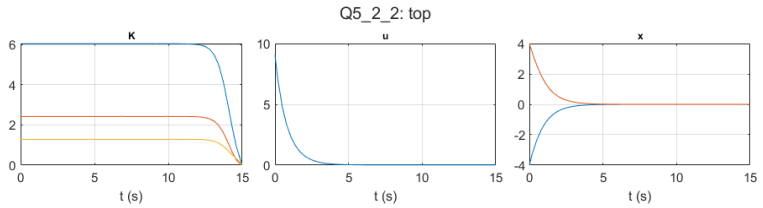
English summary of my original Persian report, which is kept as a PDF in this folder.

Example 5.2-1: first-order plant (`Q5_2_1.slx`)

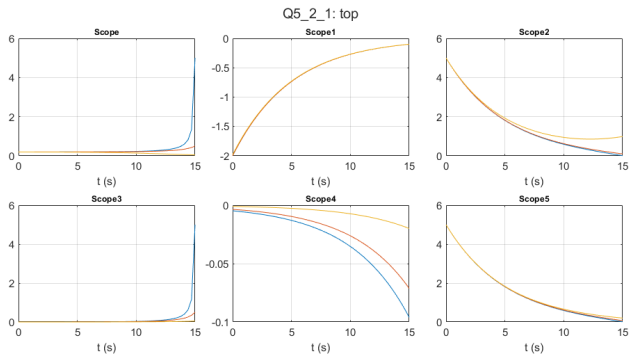
I simulated the finite-horizon LQR for a first-order plant with $a = 0.2$ (unstable) and $a = -0.2$ (stable), and plotted the Riccati gain $k(t)$, the control $u(t)$ and the state $x(t)$. With $a = 0.2$ the control is larger, because the controller has to work against the instability. Each case was run for several values of the terminal weight H. A larger H makes the control a little larger, because the controller works harder to shrink the final error that J penalises.

Example 5.2-2: second-order plant (`Q5_2_2.slx`)

The method in the course is offline: the Riccati differential equation is integrated backwards from $t_f = 15\,\text{s}$ before the simulation. `Q5_2_2_Gains.m` does this with `ode45` for K_{11}, K_{12} and K_{22} and stores them as time series. The Simulink model then applies $u(t) = -R^{-1}B^T K(t)x(t)$ to the double-integrator-type plant.



Re-run in R2023b: Riccati gains $K(t)$, control $u(t)$ and states $x(t)$.



Example 5.2-1: scopes of the six sub-models.

How to run: run `Q5_2_2_Gains.m` first, then `Q5_2_2.slx`.